Sparse Congruence-Preserving Perturbations of Dyadic Series

What gap statistics can tell us about Erdős Problem #251

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Abstract

We perturb convergent dyadic series with nonnegative integer coefficients to attain both rational and irrational sums while preserving eventual congruences and asymptotic block statistics. Nonnegative corrections on one set of upper Banach density zero attain a nondegenerate interval above the original sum. They eventually respect any prescribed bound tending to infinity and preserve coefficient and partial-sum residues for every fixed modulus, with target-independent cutoffs. We vary adjacent correction pairs with fixed ordinary totals; growing digit ranges let later weighted choices fill the gaps between present choices. Applied to prime gaps with correction bound $\log(n+3)$, this preserves cumulative growth $n \log n$ and gives vanishing total variation error for blocks of length $o(\log \log X)$ sampled on $[X, 2X)$. Cumulative positions need not be prime; irrationality of the actual prime series remains open.

1 Introduction

Let $p_0 = 2, p_1 = 3, \ldots$ be the primes and put $g_n = p_{n+1} - p_n$. Erdős asked whether

$$\Pi = \sum_{n \geq 0} \frac{p_n}{2^{n+1}}$$

is irrational [1, p. 94][2, p. 62] [3, p. 103]. Summation by parts gives $\Pi = 2 + \sum_{n \geq 0} g_n 2^{-(n+1)}$, with convergence justified in Section 4. We study which properties of the gaps survive when the latter sum is changed to a prescribed value.

The support and every modulus cutoff are fixed before the target is chosen. The interval contains rational and irrational sums with the same eventual coefficient and partial-sum residues. A block changes only when it meets the support, whereas every correction enters the dyadic sum. With polylogarithmic corrections, empirical block laws agree asymptotically at lengths $o(\log \log X)$; for prime gaps, cumulative growth remains $n \log n$. The comparison positions need not be consecutive primes.

We use a standard interval-covering argument for series with finite choices. Fridy's generalised-base lemma [5, p. 194] treats prescribed digit bounds and nonincreasing

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weights. Crmarić and Kovač [17, Lemma 4] give the finite-choice form used here, and Kovač and Tao [6, Lemma 5.1] use analogous intervals of reciprocal choices. We must arrange these choices on a sparse set while preserving cumulative congruences. To do so, we redistribute a fixed correction between two adjacent indices. The simplest useful calculation is

(e_n, e_{n+1})	$(0,6)$	$(2,4)$	$(4,2)$	$(6,0)$
$e_n + e_{n+1}$	6	6	6	6
$2^{n+2}(e_n 2^{-n-1} + e_{n+1}2^{-n-2})$	6	8	10	12.

Moving 2 from the later index to the earlier one changes the weighted sum while leaving the ordinary sum fixed. If the cumulative correction before the pair is odd, an additional 1 at $n-1$ clears its residue modulo 2. Since the four pairs have the same sum, the choice of pair leaves every later cumulative residue unchanged. We repeat this operation with nested moduli and increasingly separated pairs. The larger separations make the weighted contributions smaller, so we enlarge the range of each pair until the later contributions fill the gaps between consecutive present choices.

Throughout, $\mathbb{N} = \{0, 1, \dots\}$, intervals of indices contain integers, and log is natural unless a base is displayed. A set $S \subseteq \mathbb{N}$ has *upper Banach density zero* if

$$\lim_{H\to\infty} \sup_{u\in\mathbb{N}} \frac{|S \cap [u, u+H)|}{H} = 0.$$

For integers $X, m \geq 1$, let $\mu_{a,X,m}$ denote the distribution of $(a_n, \dots, a_{n+m-1})$ when n is uniform on $[X, 2X)$. The entries are not rescaled: we sample starting indices, not distinct block values. We use $d_{\mathrm{TV}}(\mu, \nu) = \sup_B |\mu(B) - \nu(B)|$, with the supremum over all sets of blocks.

Proposition 1.1 (sparse changes preserving congruences)**.** *Let $a : \mathbb{N} \to \mathbb{N}$ satisfy $A = \sum_{n\geq 0} a_n 2^{-(n+1)} < \infty$, let $K \in \mathbb{N}$, and let $f : \mathbb{N} \to \mathbb{R}$ tend to $+\infty$. There exist a set $S \subseteq [K, \infty)$ of upper Banach density zero and a nondegenerate interval $I \subset (A, \infty)$ such that, for every $r \in I$, there is an integer correction $e : \mathbb{N} \to \mathbb{N}$ satisfying*

$$\operatorname{supp} e \subseteq S, \qquad e_n \leq f(n) \text{ eventually}, \qquad \sum_{n\geq 0} (a_n + e_n)2^{-(n+1)} = r.$$

For each integer $q \geq 1$ there is a cutoff N_q, independent of r, such that $q \mid e_n$ and $q \mid \sum_{i<n} e_i$ for all $n \geq N_q$. For every $\varepsilon > 0$, the choice $f(n) = (\log(n+3))^{\varepsilon}$ can be made with

$$|S \cap [X, 2X)| = O_\varepsilon\left(\frac{X}{\log\log X}\right), \qquad d_{\mathrm{TV}}(\mu_{a,X,m}, \mu_{a+e,X,m}) \leq \frac{m|S \cap [X, 2X+m)|}{X}.$$

In particular the distance tends to zero for every integer-valued $m = m(X) \geq 1$ with $m(X) = o(\log\log X)$, uniformly over target values.

The cutoff may depend on q, but S and all cutoffs are fixed before r. Only e varies with the target, and its support may occupy just part of S. Infinitely many changes are essential: finitely many integer dyadic corrections add a rational number. The allowance

may grow as slowly as $\log\log(n+3)$, but it cannot be bounded. A modulus exceeding the bound would force e eventually to vanish; its constant cumulative sum would then be divisible by every positive integer. Nonnegativity forces $e = 0$. Polynomially growing coefficients satisfy the convergence hypothesis; $a_n = 2^n$ does not.

Section 2 proves the construction and Section 3 transfers the stated prime-gap properties. For the actual primes, integral tail shifts characterise rationality. Two consecutive small differences with unequal gaps give a sufficient condition for irrationality. Finite enclosures certify individual witnesses without supplying arbitrarily late ones. The companion's literature discussion contains the further comparisons, including finite subsums and other dyadic coefficient sequences.

2 Proof of Proposition 1.1

Proof. Fixed totals. Use $n_j - 1$ to cancel a cumulative residue and $n_j, n_j + 1$ for a pair with fixed ordinary sum. Varying the pair changes its weighted contribution without altering any later residue to be cancelled. Choose an initial index $n_{-1} \geq K$ and centres with $s_j = n_j - n_{j-1} \geq 4$, so the triples do not overlap. Let M_j be positive integers with $M_j \mid M_{j+1}$, and let D_j bound the digit at stage j. The spacing, moduli and digit bounds will be chosen below.

Let C_j record the ordinary correction before the jth triple. Starting from $C_0 = 0$, prescribe

$$c_j = (-C_j) \bmod M_j \quad (0 \leq c_j < M_j), \qquad C_{j+1} = C_j + c_j + M_j D_j.$$

For integers $0 \leq d_j \leq D_j$, set

$$e_{n_j-1} = c_j, \qquad e_{n_j} = M_j d_j, \qquad e_{n_j+1} = M_j(D_j - d_j), \tag{1}$$

and put $e_n = 0$ elsewhere. The two sums of a triple are

$$\sum_{r=n_j-1}^{n_j+1} e_r = c_j + M_j D_j,$$

$$\sum_{r=n_j-1}^{n_j+1} e_r 2^{-r-1} = c_j 2^{-n_j} + M_j D_j 2^{-n_j-2} + d_j M_j 2^{-n_j-2}.$$

The ordinary total contains no d_j, so the recurrence fixes every c_j in advance. In the weighted total only the last term varies: transferring M_j from $n_j + 1$ to n_j adds

$$w_j = M_j 2^{-n_j-2}.$$

After c_j has been inserted, the cumulative correction is divisible by M_j; the two entries of the pair preserve this divisibility. Fix $q \mid M_J$. At index n_J both the coefficient correction and the cumulative correction are divisible by q. At every later triple, q divides C_j and M_j, so it also divides c_j and both pair entries. The intervening corrections are zero. Hence

$$q \mid e_n, \qquad q \mid \sum_{i<n} e_i \qquad (n \geq n_J).$$

The entry c_J need not be divisible by q, which is why it is placed at $n_J - 1$, before the cutoff. The index n_J depends on q and the chosen moduli, but not on the digits or the value to be attained.

Spacing and size. The choices at stage j are $0, w_j, \dots, D_j w_j$. The covering argument will work if the later ranges total at least w_j. Taking $D_i = 2^{s_i} - 1$ compensates for separation by making each later range a scaled dyadic difference:

$$D_i w_i = (2^{n_i - n_{i-1}} - 1) M_i 2^{-n_i - 2} = M_i (2^{-n_{i-1}-2} - 2^{-n_i - 2}).$$

Because $M_i \geq M_j$ for $i > j$, replacing every later modulus by M_j gives the finite estimate

$$\begin{aligned} \sum_{i=j+1}^{J} D_i w_i &\geq M_j \sum_{i=j+1}^{J} (2^{-n_{i-1}-2} - 2^{-n_i - 2}) \\ &= w_j - M_j 2^{-n_J - 2} \qquad (J > j). \end{aligned} \tag{2}$$

No monotonicity of the weights is needed. Each entry of (1) is at most $M_j 2^{s_j}$, so the size cost of separating the stages is exponential. We must fit this product below the allowance at all three indices, while $s_j \to \infty$ and every fixed positive integer eventually divides M_j.

To allow a nonmonotone bound f, put

$$h(n) = \inf_{m \geq n} \min(f(m), m).$$

This nondecreasing lower envelope tends to infinity even when f oscillates. A bound at n_{j-1} therefore controls all three later coordinates; the cap by m ensures $e_n \leq n$ and hence summability for every digit choice. Enlarge $n_{-1} \geq K$ until $h(n_{-1}) \geq 32$, and set

$$k_j = \max\{k \geq 2 : k! 2^{k+2} \leq h(n_{j-1})\}, \qquad M_j = k_j!, \qquad s_j = k_j + 2, \qquad n_j = n_{j-1} + s_j.$$

The maximum exists: $k = 2$ is admissible because $2!2^4 = 32$, and $k!2^{k+2} \to \infty$. Since $n_j \to \infty$ and h is nondecreasing and divergent, k_j is nondecreasing and tends to infinity. Thus $M_j \mid M_{j+1}$, every fixed q eventually divides $M_j = k_j!$, and $s_j = k_j + 2 \to \infty$. At every coordinate n of the jth triple, this schedule also gives

$$0 \leq e_n \leq M_j 2^{s_j} \leq h(n_{j-1}) \leq \min(f(n), n).$$

Let $S = \bigcup_{j \geq 0} \{n_j - 1, n_j, n_j + 1\}$. For $R \geq 4$, all sufficiently late centres are at least R apart. Every interval of length H contains at most $3(H/R + 2)$ points from those triples; the earlier triples add a fixed number, independent of the interval's position. Divide by H, take the supremum over positions, then let $H \to \infty$. The limiting upper bound is $3/R$; letting $R \to \infty$ proves upper Banach density zero.

Attaining the interval. The support and congruence requirements are now settled independently of the target. Only the digits remain to be chosen. Define

$$\beta = \sum_{j \geq 0} (c_j 2^{-n_j} + D_j w_j), \qquad F_j = \sum_{i \geq j} D_i w_i.$$

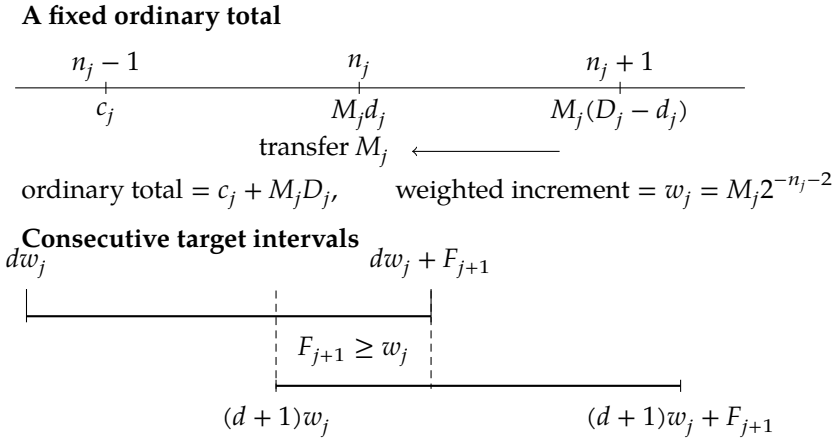


Figure 1: Redistributing M_j between adjacent entries keeps their ordinary sum fixed and increases the dyadic sum by w_j. The candidate target intervals in the lower panel cover without gaps when $F_{j+1} \geq w_j$. The sketch shows strict overlap; the proof also allows equality and establishes attainment by keeping the remainder in $[0, F_j]$ as $F_j \to 0$.

Even zero digits contribute $\beta > 0$; F_j is the maximum variable contribution from stage j onwards. The size bound makes both finite, with $F_0 > 0$ and $F_j \to 0$. The total correction is

$$\sum_n e_n 2^{-n-1} = \beta + \sum_j d_j w_j.$$

Taking $J \to \infty$ in (2) now gives

$$F_{j+1} \geq M_j 2^{-n_j-2} = w_j. \tag{3}$$

These bounds do not yet prove that every point of $[0, F_j]$ is attained. A digit d leaves a remainder in $[0, F_{j+1}]$ precisely for current remainders in

$$[dw_j, dw_j + F_{j+1}], \qquad d = 0, \dots, D_j.$$

These intervals meet or overlap by (3), and their union is $[0, F_j]$ because $F_j = D_j w_j + F_{j+1}$. Figure 1 shows the two calculations. Given $x \in [0, F_0]$, start with $\rho_0 = x$. Having chosen the first j digits, choose d_j so that

$$\rho_{j+1} = \rho_j - d_j w_j \in [0, F_{j+1}].$$

The covering just proved makes this possible at every stage. Since $0 \leq \rho_j \leq F_j$ and $F_j \to 0$, we obtain

$$x - \sum_{i=0}^{j} d_i w_i = \rho_{j+1} \longrightarrow 0.$$

Thus every $x \in [0, F_0]$ is attained, and we may take $I = (A + \beta, A + \beta + F_0)$. The target $r \in I$ determines only the digits, through $x = r - A - \beta$; it changes neither S, I nor the cutoff $N_q = n_J$ chosen using $q \mid M_J$. This last selection is the standard finite-choice

covering argument of [17, Lemma 4]. The construction above supplies its summability and overlap hypotheses while imposing the congruences and requiring the support to lie in S.

The logarithmic scale. For $f(n) = (\log(n+3))^{\varepsilon}$, the size cost 2^{s_j} leads to the iterated logarithm. We choose separations of order $\log\log n_j$ and reserve part of the allowance for growing moduli. Keep the same triples and $D_j = 2^{s_j} - 1$, and set

$$s_j = \left\lfloor \frac{\varepsilon}{2} \log_2 \log(n_{j-1}+3) \right\rfloor,$$
$$k_j = \max\{k \geq 2 : k! \leq (\log(n_{j-1}+3))^{\varepsilon/4}\},$$

with $M_j = k_j!$ and $n_j = n_{j-1} + s_j$. Start far enough out that $s_j \geq 4$ and all choices exist. The exponents $\varepsilon/4$ and $\varepsilon/2$ add to $3\varepsilon/4 < \varepsilon$, so the corrections satisfy the prescribed bound:

$$M_j 2^{s_j} \leq (\log(n_{j-1}+3))^{3\varepsilon/4} \leq (\log(n+3))^{\varepsilon}$$

at each coordinate n of the triple. After enlarging the initial index, this is also at most n. The moduli remain nested, and every fixed positive integer divides all sufficiently late moduli. Summability, the congruence induction and (2) therefore apply without change. Since $s_j \asymp_{\varepsilon} \log\log n_j$, the relevant centres near $[X, 2X)$ are separated by $\gg_{\varepsilon} \log\log X$. Each contributes at most three support points, giving

$$|S \cap [X, 2X)| = O_{\varepsilon}(X/\log\log X).$$

Choose the same uniform starting index $n \in [X, 2X)$ for both blocks. They agree when the block avoids S. A changed coordinate $t \in S \cap [X, 2X+m)$ affects only starts in $[t-m+1, t]$, at most m choices. The union bound gives

$$\begin{aligned} d_{\mathrm{TV}}(\mu_{a,X,m}, \mu_{a+e,X,m}) &\leq \mathbb{P}(e_{n+i} \neq 0 \text{ for some } 0 \leq i < m) \\ &\leq \frac{m|S \cap [X, 2X+m)|}{X}. \end{aligned}$$

For $m \leq X$, two dyadic bands cover $[X, 2X+m)$, giving $O_{\varepsilon}(m/\log\log X)$. A block of length $o(\log\log X)$ therefore avoids S with probability tending to one. The convergence is uniform over targets because the same S contains every correction support.

□

No limiting block law is assumed. A block-only test bounded in absolute value by B_0 changes its mean by at most $2B_0 d_{\mathrm{TV}}$. For such a test also depending on the starting index, the common-index coupling gives $2B_0 m|S \cap [X, 2X+m)|/X$ instead. Vanishing absolute error can still erase a rare event; it supplies neither relative-frequency control nor a comparison of complete tails. The scale $o(\log\log X)$ is sufficient, not proved optimal. The companion's Section 2 also distinguishes upper Banach density from ordinary density.

3 Application to prime gaps

The elementary bound $p_n \le 1250(n+1)^4$ suffices for convergence of the prime and gap series. We use the central-binomial argument of Erdős [19, pp. 194–196]; Appendix A of the companion derives the prime-power estimate and the bound. We may therefore apply Proposition 1.1 to the gaps.

Corollary 3.1 (a rational sum with the stated prime-gap statistics)**.** *Let $p_0 = 2, p_1 = 3, \dots$ be the primes and $g_n = p_{n+1} - p_n$. Given $K \in \mathbb{N}$ and $0 < \varepsilon \le 1$, there is $b : \mathbb{N} \to \mathbb{N}$ with rational dyadic sum such that $b_n = g_n$ for $n < K$, $b_n \ge g_n$, and $b_n - g_n \le (\log(n+3))^\varepsilon$ eventually. For every fixed positive modulus, both the coefficients and the cumulative positions eventually retain their corresponding residues. The empirical distributions of unnormalised blocks have total variation distance tending to zero for lengths $o(\log\log X)$, and for every fixed nonzero $F \in \mathbb{Z}[x_0, \dots, x_k]$,*

$$|\{n < N : F(b_n, \dots, b_{n+k}) = 0\}| = o(N).$$

Moreover, with $P_n = 2 + \sum_{i<n} b_i$,

$$0 \le P_n - p_n = O_\varepsilon\left(\frac{n(\log(n+3))^\varepsilon}{\log\log n}\right), \qquad P_n \sim n\log n.$$

These positions are not asserted to be prime.

Lean proves this for every $\varepsilon > 0$, assuming Schlage-Puchta's Lemma 4 [18] only for the nonconcentration bound and the prime number theorem only for $P_n \sim n\log n$.

Proof. Apply Proposition 1.1 to $a = g$, choose a rational $r \in I$, and fix the resulting sequence $b = g + e$. The correction, congruence and growing-block assertions follow at once. It remains to transfer polynomial nonconcentration and cumulative growth. For fixed k and nonzero $F \in \mathbb{Z}[x_0, \dots, x_k]$, a new zero requires a block meeting S, so

$$\begin{aligned}&|\{n < N : F(b_n, \dots, b_{n+k}) = 0\}| \\ &\quad \le |\{n < N : F(g_n, \dots, g_{n+k}) = 0\}| + \sum_{i=0}^{k} |S \cap [i, N+i)|.\end{aligned}$$

The first count is $o(N)$ by Schlage-Puchta's Lemma 4 [18]; the finitely many support counts are $o(N)$ because S has density zero. We count the event $F = 0$, not averages of F, so no bound on its values is needed. Since b was chosen before F, k, the same sequence satisfies every fixed-polynomial condition.

To estimate $P_n - p_n = \sum_{i<n} e_i$, first observe that $|S \cap [0, n)| = O_\varepsilon(n/\log\log n)$. Indeed, the indices below $\sqrt{n}$ contribute at most $\sqrt{n}$ points, and on $[\sqrt{n}, n)$ we sum the dyadic support bounds, whose lengths add to $O(n)$ and whose $\log\log X$ are comparable to $\log\log n$. Multiplying by the pointwise correction bound gives the stated error. For $0 < \varepsilon \le 1$ this is $o(n\log n)$, so the prime number theorem [12, Chapter 6] gives $P_n \sim n\log n$. Finally, summing nonnegative terms in the opposite order, we obtain

$$\sum_{n\ge 0} \frac{P_n}{2^{n+1}} = 2 + \sum_{i\ge 0} b_i \sum_{n>i} 2^{-n-1} = 2 + \sum_{i\ge 0} \frac{b_i}{2^{i+1}} = 2 + r \in \mathbb{Q}.$$

□

For $\varepsilon > 1$ we may use exponent 1, which gives smaller corrections and the error $O(n \log(n+3)/\log\log n)$. Thus the conclusions extend to every $\varepsilon > 0$. For each fixed $y \geq 2$, eventual congruence modulo $y!$ also gives $\gcd(P_n, y!) = 1$ once $p_n > y$. The cutoff depends on y; primality would require excluding every prime up to $\sqrt{P_n}$ at that same index. Even prime-valued positions could omit intervening primes.

Land's draft [8, Theorem 2] proves conditional irrationality under a uniform Hardy–Littlewood hypothesis of the kind formulated by Kuperberg [7, Conjecture 1.3]. Ringer's draft [16] obtains conditional normality. Their quantitative prime-pattern hypotheses are stronger than the absolute block comparison above and are not assumed here. The companion's Section 3 compares the hypotheses and sampling conventions.

4 The prime series and its actual tails

We now study complete tails of the actual prime gaps. Put $G = \sum_{n\geq 0} g_n 2^{-(n+1)}$. The polynomial bound from Section 3 proves convergence and makes the endpoint in finite summation by parts vanish.

Theorem 4.1 (prime-to-gap identity)**.** *The actual prime and gap series satisfy* $\Pi = 2 + G$. *Their complete tails*

$$T_N = \sum_{j\geq 1} g_{N+j} 2^{-j}$$

satisfy $T_0 = 2G - 1$ *and* $T_{N+1} = 2T_N - g_{N+1}$. *Thus* Π, G *and* T_0 *have the same rationality status.*

Proof. Finite summation by parts gives

$$\sum_{i=0}^{n} \frac{p_i}{2^{i+1}} = 2 + \sum_{i=0}^{n-1} \frac{g_i}{2^{i+1}} - \frac{p_n}{2^{n+1}}.$$

Since $p_n = O((n+1)^4)$, the endpoint tends to zero and we obtain $\Pi = 2 + G$. We obtain the tail recurrence by splitting off the first term of T_N; removing $g_0 = 1$ from G gives $T_0 = 2G - 1$. □

Tao noted this reduction in the problem's forum discussion [15, 7 October 2025].

Corollary 4.2 (exact irrationality reformulation)**.** *The prime-value dyadic series is irrational if and only if the prime-gap dyadic series is. Both series converge by the polynomial bound proved above; neither side is proved irrational.*

Proof. Adding the rational number 2 preserves rationality and irrationality. □

The index N is the last omitted coefficient, so T_N starts with $g_{N+1}/2$. Rescaling distinguishes these tails from the remainders of G. The recurrence also admits an extra term $C2^N$; the vanishing boundary condition below excludes it.

Lemma 4.3 (the boundary condition for the tail recurrence)**.** *Let* $\sum_{j\geq 1} |a_j| 2^{-j} < \infty$ *and* $U_{N+1} = 2U_N - a_{N+1}$. *Then* $U_N = \sum_{j\geq 1} a_{N+j} 2^{-j}$ *for every* N *if and only if* $2^{-N} U_N \to 0$.

Proof. Iteration gives $2^{-N}U_N = U_0 - \sum_{j=1}^{N} a_j 2^{-j}$. The limit is zero exactly when U_0 is the full series. Subtracting its first N terms and multiplying by 2^N then gives the tail formula. Conversely the formula implies $2^{-N}U_N = \sum_{k>N} a_k 2^{-k} \to 0$ by absolute convergence. □

4.1 Integral shifts

For $U_{N+1} = 2U_N - a_{N+1}$ with $a_n \in \mathbb{Z}$, write $D_h(N) = U_{N+h} - U_N$. Modulo integers, each step doubles, so an integral shift is a return to the same fractional part. For rational U_0, the power of two in its reduced denominator determines when returns of positive length can begin; the odd part determines their lengths.

Theorem 4.4 (exact rationality classification)**.** *For a real integer-coefficient recurrence, the following are equivalent: $U_0 \in \mathbb{Q}$; $D_h(N) \in \mathbb{Z}$ for some $h \geq 1, N \geq 0$; and, for some fixed $h \geq 1$, $D_h(N) \in \mathbb{Z}$ at every sufficiently large N. More precisely, if $U_0 = u/(2^s d)$ is in lowest terms, with d odd, then*

$$\operatorname{den}(U_N) = 2^{\max(s-N,0)}d, \qquad D_h(N) \in \mathbb{Z} \iff N \geq s \text{ and } d \mid 2^h - 1 \quad (h \geq 1).$$

Consequently U_0 is irrational exactly when every positive shift is nonintegral at every index, equivalently when for each fixed $h \geq 1$ there are arbitrarily late nonintegral shifts.

Proof. Iteration gives

$$U_N - 2^N U_0 \in \mathbb{Z}, \qquad D_h(N) - 2^N(2^h - 1)U_0 \in \mathbb{Z}.$$

An integral shift with $h \geq 1$ makes the nonzero integer multiple $2^N(2^h - 1)U_0$ integral, and therefore forces $U_0 \in \mathbb{Q}$. Conversely, write $U_0 = u/(2^s d)$ in lowest terms. Multiplication by 2^N cancels exactly $\min(N, s)$ powers of two from the denominator. The remaining factor $2^h - 1$ is odd, so the second relation is integral exactly when $N \geq s$ and $d \mid 2^h - 1$. For $d > 1$, Euler's congruence provides such a shift $h = \varphi(d)$, and for $d = 1$ we take $h = 1$. This shift works for every $N \geq s$. Negating the pointwise and eventual criteria gives the two assertions about irrationality. □

The classification applies to any integer-coefficient recurrence; Lemma 4.3 is needed only to identify complete tails. The periodicity is of fractional parts, not necessarily tail values. For the positive even coefficients $2, 4, 2, 4, \ldots$ starting at $n = 1$, the tails alternate between $8/3$ and $10/3$. Length-1 shifts are nonintegral, but length-2 shifts vanish: one shift length cannot suffice.

5 Small tail differences and finite tests

Fix $h \geq 1$ and put $D_N = T_{N+h} - T_N$ and $\delta_N = g_{N+h+1} - g_{N+1}$. Then $D_{N+1} = 2D_N - \delta_N$. Both gap indices are positive, so neither involves the exceptional odd gap $g_0 = 1$; hence δ_N is even. Two integral differences of absolute value less than one must both vanish, forcing $\delta_N = 0$. Evenness yields the sharper signed test below.

Proposition 5.1 (two consecutive differences of absolute value less than one)**.** *Let $D, D' \in \mathbb{R}$, $\delta \in 2\mathbb{Z}$ and $D' = 2D - \delta$. The conditions $|D| < 1$, $|D'| < 1$ and $\delta \neq 0$ hold exactly when, for some $s \in \{-1, 1\}$,*

$$\delta = 2s, \qquad \tfrac{1}{2} < sD < 1.$$

In that case $sD' \in (-1, 0)$ and both D and D' are nonintegral.

Proof. The bounds on D, D' imply $-3 < \delta < 3$, so a nonzero even δ equals $2s$ with $s \in \{-1, 1\}$. Substituting in $D' = 2D - 2s$ gives $1/2 < sD < 1$. Conversely this interval gives $sD' = 2sD - 2 \in (-1, 0)$. □

The endpoints cannot be included: when $\delta = 2$, $D = 1/2$ gives $D' = -1$, and $D = 1$ already violates $|D| < 1$. The zero triple $D = D' = \delta = 0$ shows why the mismatch is also required.

Problem 5.2 (two small tail differences at arbitrarily large indices)**.** For every integer $h \geq 1$ and every cutoff N_0, exhibit $N \geq N_0$ with

$$|T_{N+h} - T_N| < 1, \quad |T_{N+h+1} - T_{N+1}| < 1, \quad g_{N+h+1} \neq g_{N+1}. \tag{4}$$

Theorem 4.4 requires nonintegrality, not smallness; (4) is only a sufficient condition. All three requirements must hold at the same arbitrarily late indices. Separate infinite sets need not intersect.

By Proposition 5.1, such an index must have $\delta_N = \pm 2$. Schlage-Puchta's lemma [18, Lemma 4], applied to $x_h - x_0 \pm 2$, shows that these eligible indices already have density zero. Density zero is compatible with arbitrarily late witnesses; it supplies no lower bound for their occurrence. The absolute block comparison in Proposition 1.1 likewise gives no relative-frequency estimate for this rare event.

5.1 Finite separation from the integers

An enclosure certifies nonintegrality only when it misses every integer. Choose $M(n) \geq g_n$ with $\sum_n M(n)2^{-n} < \infty$, and write

$$S_{h,N,L} = \sum_{j=1}^{L} (g_{N+h+j} - g_{N+j})2^{-j}, \qquad R_{h,N,L}(M) = \sum_{j>L} (M(N+h+j) + M(N+j))2^{-j}.$$

Splitting the complete tails after L terms gives

$$D_N = S_{h,N,L} + 2^{-L} D_{N+L}, \qquad |D_N - S_{h,N,L}| \leq R_{h,N,L}(M).$$

The uncomputed term is a rescaled later complete tail difference with the same shift h. An arbitrary real majorant gives an enclosure, but need not give a computable radius.

For the actual prime gaps, one computable choice is $M(n) = 1250(n+2)^4$. For $P(x) = x^4 + 8x^3 + 36x^2 + 104x + 150$, the identity $2P(x) = (x+1)^4 + P(x+1)$ telescopes to $\sum_{j=1}^{J} (x+j)^4 2^{-j} = P(x) - 2^{-J} P(x+J)$. The endpoint vanishes as $J \to \infty$, giving

$$R_{h,N,L}(M) = 1250\, 2^{-L} (P(N+h+L+2) + P(N+L+2)).$$

Proposition 5.3 (finite separation criterion). *If for every $h \geq 1$ and every cutoff N_0 there are $N \geq N_0, L \geq 1$ with*

$$\operatorname{dist}(S_{h,N,L}, \mathbb{Z}) > R_{h,N,L}(M), \tag{5}$$

then Π is irrational.

Proof. The enclosure gives $\operatorname{dist}(D_N, \mathbb{Z}) \geq \operatorname{dist}(S_{h,N,L}, \mathbb{Z}) - R_{h,N,L}(M) > 0$. Thus each fixed positive shift is nonintegral at arbitrarily large indices, and Theorem 4.4 applies. □

Remark (what arbitrary truncation does not gain). Whenever $|D - S_L| \leq R_L \to 0$, nonintegrality of D is equivalent to $\operatorname{dist}(S_L, \mathbb{Z}) > R_L$ for some L. Indeed, for $\delta = \operatorname{dist}(D, \mathbb{Z}) > 0$, take $R_L < \delta/2$ and use the 1-Lipschitz property of distance. Unrestricted depth therefore gives an exact reformulation, not an additional analytic estimate. At a prescribed logarithmic depth one must also control the margin from the integers as N varies; a shrinking remainder alone does not do so. The companion, Appendix D.2, gives an explicit example distinguishing refinement at a fixed value from enclosures of changing values.

A nonintegral value need not lie in either signed interval of Proposition 5.1. To certify (4), suppose $\delta_N = 2s$, with $s \in \{-1, 1\}$, and enclose D_N using integers A, Q, B with $Q > 0$, $B \geq 0$ and $|QD_N - A| \leq B$. Then $sD_N \in [(sA - B)/Q, (sA + B)/Q]$. Placing its lower endpoint above $1/2$ and its upper endpoint below 1 gives, respectively,

$$2sA - Q > 2B, \qquad Q - sA > B.$$

We retain the signed numerator: reduction modulo Q preserves distance to $Q\mathbb{Z}$ but loses the enclosure's location. The gap difference fixes s, so it cannot be chosen to fit the enclosure. At $h = 1, N = 2, L = 40$, one has $\delta_2 = g_4 - g_3 = -2$ and

$$Q = 2^{40}, \qquad A = -662838684750, \qquad B = 11764181250, \qquad s = -1.$$

Here $A/Q = S_{1,2,40}$ and $B/Q = R_{1,2,40}(M)$, and the two strict margins are 202637379224 and 424908761776. The companion's integer certificates give the prime-index calculation and remainder derivation. This finite pair leaves the quantifiers in Problem 5.2 unresolved.

6 Further consequences and questions

We can place the interval in Proposition 1.1 inside $(A, A + \eta)$ for any $\eta > 0$. Choose $K' \geq K$ such that $(K' + 1)2^{-K'} < \eta$, and start the construction after K'. The bound $e_n \leq n$ from the proof gives

$$0 < \sum_n e_n 2^{-(n+1)} \leq (K' + 1)2^{-K'} < \eta.$$

Thus rational and irrational targets remain available arbitrarily close to A, with the prescribed prefix unchanged. The estimate controls the dyadic change, not the ordinary total.

6.1 Nonperiodic coefficients

Rationality forces eventual periodicity of the binary digits obtained after carrying, not of the original integer coefficients. For example,

$$\sum_{i=0}^{n-1} \frac{i-1}{2^{i+1}} = -\frac{n}{2^n} \longrightarrow 0.$$

Its coefficients are unbounded and its first coefficient is -1. We obtain a positive even example by taking $c_n = 2(n^2+4n+2)$ and $U_n = 2(n+4)^2$. Direct substitution gives $U_{n+1} = 2U_n - c_{n+1}$, while the polynomial growth gives $2^{-n}U_n \to 0$. Lemma 4.3 therefore identifies U as the complete tail and yields

$$\sum_{j\geq 1} c_j 2^{-j} = 32, \qquad \sum_{n\geq 0} c_n 2^{-(n+1)} = 18.$$

The coefficients increase strictly and $c_{n+1} - c_n = 4n + 10$ is never ± 2. The companion's Section 8 gives further examples, including a bounded nonperiodic sequence in Appendix D. None of these examples constructs consecutive primes.

Proposition 6.1 (prime gaps do not become periodic)**.** *For every positive h, the actual consecutive-prime-gap sequence is not eventually periodic with period h.*

Proof. If the gaps were eventually periodic, they would be bounded. For each $m \geq 2$, however, the $m-1$ integers $m!+2, \dots, m!+m$ are composite. The consecutive primes on either side therefore have a gap of at least m, contradicting boundedness. □

The results on bounded prime gaps [13], clusters of each fixed size [11, 20], and extreme large gaps [4] do not give the simultaneous inequalities in (4). This remains so with Polymath's bound $\liminf_n g_n \leq 246$ [20, Theorem 1.4(i)]. For fixed h and $\varepsilon > 0$, taking $L = \lceil (4+\varepsilon)\log_2(N+2) \rceil$ in the polynomial remainder gives an error $O_{h,\varepsilon}(N^{-\varepsilon})$. The finite sum must still clear the relevant boundary by more than this error to supply a witness for Problem 5.2. The perturbation theorem rules out no argument using additional primality information or quantitative rare-event estimates.

7 Verification and reproducibility

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The verification record gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded. The linked Lean declarations use Lean 4 [9] and mathlib [10]. The formal proof of Proposition 1.1 changes one coordinate at each stage, whereas the proof here uses triples. Both choose the set containing the supports, the interval and the congruence cutoffs before the value of the sum. For Corollary 3.1, the formal statement covers every $\varepsilon > 0$ and assumes Schlage-Puchta's lemma only for nonconcentration and the prime number theorem only for the asymptotic $P_n \sim n \log n$.

The source links record formal support for the linked statements, rather than verification of the different printed proofs. The companion's Appendix G retains the complete source index at this note's original pin, including the finite telescoping identities and their nonperiodicity consequences. Its Section 3 retains the broader subsum and automatic-sequence comparisons.

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Verification concordance

Statement	Formal proof	Statement	Formal proof
Proposition 1.1	Lean · Comparator	Corollary 3.1	Lean†
Theorem 4.1	Lean · Comparator	Corollary 4.2	Lean · Comparator
Lemma 4.3	Lean · Comparator	Theorem 4.4	Lean · Comparator
Proposition 5.1	Lean · Comparator	Proposition 5.3	Lean · Comparator
Proposition 6.1	Lean · Comparator		